

DYLAN S. DAVIS, Ph.D.

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Adjunct Assoc. Research Scientist
Columbia Climate School
Columbia University

Visiting Scholar
Department of Geography
Binghamton University

Geospatial Projects Manager
Research & Development
XTS Inc.

EDUCATION

- 2018 – 2022 **Ph.D.** in Anthropology, Pennsylvania State University
Dissertation Title: *Living with change: An archaeological study of human settlement patterns as environmental adaptations in Late Holocene Madagascar*. Advisor: Kristina G. Douglass
- 2017 – 2018 **M.A.** in Anthropology, Binghamton University, New York
Advisor: Carl P. Lipo
- 2014 – 2017 **B.S.** Anthropology & **B.A.** Geography, Binghamton University, New York (*summa cum laude*)
Concentrations: Archaeology, Computer Applications in Human Environmental Analysis

ACADEMIC APPOINTMENTS

- 2026 – Pres.: Visiting Scholar, Department of Geography, Binghamton University
- 2025 – Pres.: Adjunct Associate Research Scientist, Columbia Climate School, Columbia University
- 2024 – 2025: Postdoctoral Research Scientist, Lamont-Doherty Earth Observatory, Columbia Climate School, Columbia University
- 2022 – 2024: National Science Foundation SBE Postdoctoral Research Fellow, Columbia University
- 2021 – 2022: Graduate Lecturer, The Pennsylvania State University
- 2020 – 2021: Graduate Research Assistant, The Pennsylvania State University
- 2019 – 2020: Graduate Research Fellow, NASA Pennsylvania Space Grant Consortium
- 2018 – 2021: Teaching Assistant, The Pennsylvania State University

INDUSTRY RESEARCH APPOINTMENTS

- 2025 – Present: **Geospatial Projects Manager**, Xcellent Technology Solutions, Warrenton VA.

RESEARCH INTERESTS

- | | | |
|--------------------------------------|--|------------------------------|
| – Human-environment systems dynamics | – geospatial analysis | – environmental archaeology |
| – remote sensing and GIS | – complex adaptive systems | – ethical research practices |
| – climate and social organization | – human adaptations to hypervariable climate | – landscape archaeology |
| – machine learning and AI | – geophysical applications in archaeology | – island and coastal studies |
| | | – environmental modeling |

EDITED WORKS

(*denotes equal contribution)

- In Prep.* M. Cavalleri, D.S. Davis (editors). “AI-Aided Discovery in Archaeology: Applications of ML/AI to Archaeological Data Collection, Analysis, and Interpretation [Special Issue]” ***Advanced Intelligent Discovery***. John Wiley & Sons.

- 2025 **D. S. Davis***, Hackel, J.*, L. N. Phelps*, E. Razanatsoa*, T. Rasolondrainy*, and G. Tiley* (editors) "Madagascar's grassy biomes: from Holocene to Anthropocene [Special Issue]." **Plants, People, Planet**. John Wiley & Sons.
- 2021 **Davis, D. S.**, and M. C. Sanger (editors). "Ethical Practice in Remote Sensing and Geophysical Archaeology [Special Issue]." **Archaeological Prospection**. 28(3):269-399. John Wiley & Sons.

PEER REVIEWED PUBLICATIONS (*student author)

Gold Open Access  Green Open Access 

- In Press* **Davis, D. S.** and K. Douglass. "Reframing Remote Sensing in Africanist Archaeology: Ethics, Power, and Participatory Practice". In: McCall, Álvarez Larrain, A., M. N. A. Cho, & M. K., Panek, J., (eds.), *Participatory mapping and PPGIS: Mapping with People*. pp. 55-72. Edward Elger.
- 2026 **D. S. Davis** and C. P. Lipo. "The Future of Archaeological Prospection." **Archaeological Prospection**. In Press. <https://doi.org/10.1002/arp.70046>
- 2025 **D. S. Davis**, G. Manahira, B. Tucker, and K. Douglass. "Traditional land-use is integral to ecological resilience on SW Madagascar." **Scientific Reports**. 15(1): 31160 
- 2025 Hackel, J., **D. S. Davis**, T. Rasolondrainy, E. Razanatsoa, G. Tiley, and L. N. Phelps. "Madagascar's grassy biomes, from Holocene to Anthropocene." **Plants, People, Planet**. In Press. 
- 2025 Phelps, L. N., **D.S. Davis**, J. Chen, S. Monroe, C. Anyanwu*, and K. Douglass. "Africa-wide diversification of livelihoods strategies: Isotopic insights into Holocene human adaptations to climate change." **One Earth**. 8(6): 101304. 
- 2025 Phelps, L. N., E. Razanatsoa, **D.S. Davis**, J. Hackel, T. Rasolondrainy, G. Tiley, D. Burney, R. Cox, L. Godfrey, G.P. Hempson, S. Hixon, T. Andermann, S. Andriambololonera, L.R. Andriamiarisoa, A. Antonelli, G. Besnard, C. Birkinshaw, W. Bond, L. Chikhi, V. Fernández-García, L. Gillson, S. Haberle, J. Hansford, G.S. Joseph, C.A. Kull, C.L. Mangut*, R. Marchant, V. Montade, K.V. Pham*, D. Rabehevitra, U. Radespiel, J. Raharimampionona, M.T. Rajaonah, N. Rakotomalala, T. Ramiadantsoa, B.A. Ramiandrisoa, H.L.T. Ranarijaona, T. Randriamboavonjy, F. Randrianarimanana, F. Randriatsara, J. Ratsirarson, A.H.I. Razafimanantsoa, J. Salmona, K. Samonds, N. Scroxton, C. Seymour, T. Steffens, H. Teixeira, N.R.G. Voarintsoa, P.O. Waeber, L. Wilmé, A.D. Yoder, E.C. Fisher, C. Solofondranohatra, T. van Elst, B. Crowley, K. Douglass, M.S. Vorontsova, C.E.R. Lehmann. "Advancing transdisciplinary research on grassy biomes to support resilience in tropical ecosystems and livelihoods." **Ecological Monographs**. 95(2): e70011. 
- 2024 **Davis, D. S.**, A. Domic, G. Manahira, and K. Douglass. "Geophysics elucidate long-term socio-ecological dynamics of foraging, pastoralism, and mixed subsistence strategies on SW Madagascar." **Journal of Anthropological Archaeology**. 75: 101612. 
- 2024 **Davis, D. S.**, B. Tucker, Tsiazonera, R. Justome, Z. M. Chrisostome, B. V. Pierre, A. Domic, L. Phelps, A. Ibiroba*, C. Magnut*, C. E. Klehm, and K. Douglass. "Archaeology, Ethnography, and Geosciences

- reveal central role of traditional lifeways in shaping Madagascar's dry forests." *Journal of Social Archaeology*. 24(3):221-245. 
- 2024 **Davis, D. S.**, R. J. DiNapoli, G. Pakarati, T. L. Hunt, and C. P. Lipo. "Island-wide characterization of agricultural production challenges the demographic collapse hypothesis for Rapa Nui (Easter Island)." *Science Advances*. 10(25):eado1459. 
- 2024 **Davis, D. S.**, Cook-Hale, J. W., N. Hale, T. Z. Johnston*, and M. C. Sanger. "Bathymetric lidar and semi-automated feature extraction assist underwater archaeological surveys." *Archaeological Prospection*. 31(2):171-186. 
- 2024 Alders, W., **D. S. Davis**, & J. J. Haines. "Archaeology in the fourth dimension: studying landscapes with multitemporal PlanetScope satellite data." *Journal of Archaeological Method and Theory*. 31:1588–1621. 
- 2024 **Davis, D. S.**, G. Manahira, F. Lahiniriko, V. Andriankaja, T. L. Carnat, Z. M. Chrisostome, M. B. J. Clovis, F. Fenomanana, L. Hubertine, R. Justome, H. Léonce, A. J. Yves, R. Roi, F. Victorian, V. Voahirana, R. Flerita, Z. M. Chrisostome, and K. Douglass. "Evidence of lithic blade technology in southwest Madagascar." *The Journal of Island and Coastal Archaeology*. 19(3):649–654. 
- 2024 **Davis, D. S.** "Past, present, and future of complex systems theory in archaeology." *Journal of Archaeological Research*. 32(4):549–596. 
- 2023 **Davis, D. S.**, T. Rasolondrainy, G. Manahira, V. Andriankaja, L. Hubertine, R. Justome, F. Lahiniriko, H. Léonce, R. Roi, F. Victorian, M. B. J. Clovis, V. Voahirana, T. L. Carina, A. J. Yves, Z. M. Chrisostome, B. Manjakahery, and K. Douglass. "Social networks as risk-mitigation strategies in south-west Madagascar." *Antiquity*. 97(395):1296-1312. 
- 2023 Quintus, S., **D. S. Davis**, and E. E. Cochrane. "Evaluating Mask R-CNN Models to Extract Terracing across Oceanic High Islands: an example from Sāmoa." *Archaeological Prospection*. 30(4):477-492. 
- 2023 Cook-Hale, J.[‡], **Davis, D. S.**[‡], and M. C. Sanger. "Evaluating the archaeological efficacy of bathymetric LiDAR across oceanographic contexts: A case study from Apalachee Bay, Florida." *Heritage*. 6(2):928-954. [‡]Co-first authors 
- 2022 Phelps, L. N., N. Andela, M. Gravey, **D. S. Davis**, C. A. Kull, K. Douglass, K., and C. E. R. Lehmann. "Madagascar's fire regimes challenge global assumptions about landscape degradation." *Global Change Biology*. 28(23):6944-6960. 
- 2021 **Davis, D. S.**, and J. Lundin. "Locating Charcoal Production Sites in Sweden Using Hydrological Algorithms and Deep Learning." *Remote Sensing*. 13(18):3680. 
- 2021 **Davis, D. S.**, and K. Douglass. "Remote Sensing Reveals Lasting Legacies of Land-Use by Small-Scale Communities in the southwestern Indian Ocean." *Frontiers in Ecology and Evolution*. 9: 689399. 

- 2021 **Davis, D. S.**, G. Caspari, M. C. Sanger, and C. P. Lipo. "Using Deep Learning to Detect Rare Archaeological Features: A Case from Coastal South Carolina, USA." *Society of Exploration Geophysicists Technical Program Expanded Abstracts*. 3265-3269. 
- 2021 **Davis, D. S.**, and M.C. Sanger. "Ethical Challenges in the Practice of Remote Sensing and Geophysical Archaeology." *Archaeological Prospection*. 28(3):269-276. 
- 2021 **Davis, D. S.**, G. Caspari, C. P. Lipo, and M. C. Sanger. "Deep Learning Reveals Extent of Archaic Native American Shell Ring Building Practices." *Journal of Archaeological Science*. 132:105433. 
- 2021 **Davis, D. S.** "Theoretical repositioning of automated remote sensing archaeology: Shifting from features to ephemeral landscapes." *Journal of Computer Applications in Archaeology*. 4(1): 94-109. 
- 2021 **Davis, D. S.**, D. C. Buffa, T. Rasolondrainy, E. Creswell, C. Anyanwu, A. Ibirogba, C. Randolph, A. Ouarghidi, L. N. Phelps, F. Lahiniriko, Z. M. Chrisostome, G. Manahira and K. Douglass. "The aerial panopticon and the ethics of archaeological remote sensing sacred cultural spaces." *Archaeological Prospection*. 28(3):303-318. 
- 2021 **Davis, D. S.**, K. E. Seeber, and M. C. Sanger. "Addressing the problem of disappearing cultural landscapes in archaeological research using multi-scalar survey." *The Journal of Island and Coastal Archaeology*. 16(2-4): 524-540. 
- 2020 **Davis, D. S.**, R. J. DiNapoli, and K. Douglass. "Integrating point process models, evolutionary ecology, and traditional knowledge improves landscape archaeology: A case from Southwest Madagascar." *Geosciences*. 10(8): 267. 
- 2020 **Davis, D. S.** "Defining what we study: The contribution of machine automation in archaeological research." *Digital Applications in Archaeology and Cultural Heritage*. 18: e00152. 
- 2020 **Davis, D. S.**, R. J. DiNapoli, M. C. Sanger, and C. P. Lipo. "The integration of lidar and legacy datasets provides improved explanations for the spatial patterning of shell rings in the American Southeast." *Advances in Archaeological Practice*. 8(4):361-375. 
- 2020 **Davis, D. S.**, D. C. Buffa, and A. C. Wroblewski. "Assessing the utility of open-access bathymetric data for shipwreck detection in the United States." *Heritage*. 3(2): 364-383. 
- 2020 **Davis, D. S.** "Geographic disparity in machine intelligence approaches for archaeological research." *Remote Sensing*. 12(6): 921. 
- 2020 **Davis, D. S.**, and K. Douglass. "Aerial and Spaceborne Remote Sensing in African Archaeology: A Review of Current Research and Potential Future Avenues." *African Archaeological Review*. 37(1): 9-24. 

- 2020 **Davis, D. S.**, V. Andrianakaja, T. L. Carina, Z. M. Chrisostome, C. Colombe, F. Fenomanana, L. Hubertine, R. Justome, F. Lahiniriko, H. Léonce, G. Manahira, B. V. Pierre, R. Roi, P. Soafiavy, F. Victorian, V. Voahirana, B. Manjakahery, and K. Douglass. "Satellite-based remote sensing rapidly reveals extensive record of Holocene coastal settlement on Madagascar." *Journal of Archaeological Science*. 115: 105097. 
- 2020 **Davis, D. S.** "Studying human responses to environmental change: Trends and trajectories of archaeological research." *Environmental Archaeology*. 25(2): 367-380. 
- 2019 **Davis, D. S.** "Object-based image analysis: a review of developments and future directions of automated feature detection in landscape archaeology." *Archaeological Prospection*. 26(2): 155-163. 
- 2019 **Davis, D. S.**, C. P. Lipo, and M. C. Sanger. "A comparison of automated object extraction methods for earthwork feature identification in South Carolina." *Journal of Archaeological Science: Reports*. 23: 166-177. 
- 2019 **Davis, D. S.**, M. C. Sanger, and C. P. Lipo. "Automated mound detection using LiDAR and object-based image analysis in Beaufort County, SC." *Southeastern Archaeology*. 38(1): 23-37. 
- 2017 **Davis, D. S.** "The Applicability of Short-Wave Infrared (SWIR) Imagery for Archaeological Landscape Classification on Rapa Nui (Easter Island), Chile." *Alpenglow* 3(1): 7. 

MANUSCRIPTS IN REVIEW

Davis, D. S., and K. Douglass

n.d. Landscape Dynamics and Settlement Patterns in Southwest Madagascar since the Late Holocene. In: C. Radimilahy, K. Douglass, and Z. Crossland (eds.), *The Malagasy World*. Routledge Handbooks Online: Routledge.

D. S. Davis

n.d. The Importance of Archaeological Land Use Legacies for NASAs Earth Science Directorate. To be submitted to B. Odom and K.E. Herndon (eds.), *Symposium Proceedings: NASA and Archaeology from Space*. University of Colorado Press.

D. S. Davis

n.d. Toward a locally deployed large language model pipeline for archaeological data exploration. Submitted to *Journal of Computer Applications in Archaeology*.

D. S. Davis, Phelps, L. N., P. Faina, T. Ramiadantsoa, G. Manahira, R. Justome, F. Lahiniriko, H. Léonce, K. Douglass

n.d. Where the past lies: Mapping Archaeological Potential and Heritage Risk Across Madagascar. Submitted to *Journal of Island and Coastal Archaeology*.

K. Agiadi, et al.

n.d. The Q-MARE database of climate and pre-industrial human impacts on marine ecosystems (version 2.1). Submitted to *Earth System Science Data*.

Reprinted Publications

- 2022 **Davis, D. S.**, and K. Douglass. "Aerial and Spaceborne Remote Sensing in African Archaeology: A Review of Current Research and Potential Future Avenues." In: Gokee, C., Klehm, C. (eds) *Spatial Approaches in African Archaeology*. pp. 9-24. Springer, Singapore. Reprinted from: Davis, D. S., and K. Douglass (2020) Aerial and Spaceborne Remote Sensing in African Archaeology: A Review of Current Research and Potential Future Avenues. *African Archaeological Review*. 37(1): 9-24.

Book Reviews

- 2023 **Davis, D. S.** "Review of Human Behavioral Ecology and Coastal Environments, edited by H. B. Thakar and C. Flores Fernandez." *American Antiquity*. 88(4): 588-589.

MANUSCRIPTS IN PREPARATION

Phelps, L. N., P. Faina, **D. S. Davis**, C. Mangut, A. Ibirogbra, C. Randolph, G. Manahira, R. Justome, F. Lahiniriko, H. Léonce, K. Douglass

n.d. Co-Production Strategies for Equitable Environmental Research: A case from Madagascar [Working Title]. To be submitted to **TBD**.

D. S. Davis, S. W. Hixon, A. Domic, P. Faina, K. Douglass, and W. D'Andrea,

n.d. Molecular biomarkers clarify livelihood practices in SW Madagascar over the last millennium. To be submitted to ***Proceedings of the National Academy of Sciences***.

C. P. Lipo, **D. S. Davis**, R. J. DiNapoli

n.d. Toward Label-Free Archaeological Prospection: Training-Free Detection and Zero-Shot Interpretation of Pre-European Earthworks from Public LiDAR in the Yazoo Basin and Coastal South Carolina. To be submitted to ***Archaeological Prospection***.

F. Ludlow, et al.

n.d. 4-OCEANS Historical Marine Extraction Paper [Title TBD]. To be submitted to TBD.

PUBLICATIONS FOR GENERAL PUBLIC (*Invited Submission)

- 2022 Randolph, C., **D. Davis**, C. Anyanwu, K. Pham, D. Buffa, E. Creswell, A. Ibirogbra, and K. Douglass Indigenous knowledge is key to a sustainable future. ***Penn State Institutes of Energy and the Environment Blog***.

- 2021* **Davis, D.**, D. Buffa, and A. Wroblewski. Automatic Shipwreck Detection in Bathymetry Data. ***Hydro International***. 25(1): 10-13.

FUNDING ACQUISITION

RESEARCH GRANTS – TOTAL AWARDED (approximate): **\$941,241**

External Funding: **\$865,849** | Internal Funding: **\$69,209** | Travel Funding: **\$4,454**

(CU = Columbia University, PSU = Penn State, BU = Binghamton University)

External Research Funding
(* denotes PhD student mentee)

- **European Cooperation in Science and Technology (eCOST)** – Action Grant (Secondary Proposer, PI G. Gattiglia), *Managing Artificial Intelligence in Archaeology* (Proposal ID: CA23141) €520,000 2024-2028
- **NSF Spatial Archaeometry Research Collaborations** – Residency Fellow, Center for Advanced Spatial Technologies (CAST), University of Arkansas \$1,000 2024
- **German Aerospace Center (DLR)** – TanDEM-X Mission DEM Scientific Research Proposal (Co-I, PI A. Ibirogba*): *Historical Archaeology of the Badagry Region, Lagos State, Nigeria* (Proposal ID: DEM_GEOL3906) 12000 km² of data 2023-2024
- **National Science Foundation (NSF)** – SBE Postdoctoral Research Fellowship (PI): *Long-term effects of socioeconomic activity on ecological stability* (Award No. SMA-2203789) \$143,000 2022-2024
- **American Philosophical Society** – Supplemental COVID-19 Research Support Grant (PI) \$1,000 2021
- **National Geographic Society** – Enduring Impacts RFP Grant (co-PI, PI K. Douglass): *Enduring Impacts of Climate and People on the Landscape of Southwest Madagascar* (Award No. NGS-77912R-21) \$79,995 2021-2023
- **National Science Foundation (NSF)** – Doctoral Dissertation Research Improvement Grant: *The Role of Mobility in Human-Environment Interactions* (Award No. BCS-2039927) \$31,556 2021-2023
- **NSF Spatial Archaeometry Research Collaborations** – Data and Analytics Grant (PI) – *Quantifying pastoralist niche construction on southwest Madagascar using multi-sensor remote sensing (1600 – Present)* \$12,381 2021-2022
- **NASA Pennsylvania Space Grant Consortium** – Mini-Grant (PI) *Satellites, settlements, and environment: studying past mobility patterns and their environmental contexts in Southwest Madagascar* \$4,995 2020-2021
- **Sigma Xi** – Grant in Aid of Research (PI) *Studying the effects of environmental change on human settlement Behavior in Southwest Madagascar* [Grant No. G2020031599752491] \$800 2020-2021
- **Microsoft** – AI for Earth Azure Compute Credit Grant (co-PI, PI Kristina Douglass): *Harnessing the power of automated archaeological remote sensing in Holocene SW Madagascar* \$15,000 2020-2021
- **Explorers Club** – Mamont Scholar Grant (PI) *Understanding Human-Environmental Interaction in Coastal Southwest Madagascar (ca. 10,000 – 1000 BP)* \$2,000 2020-2021
- **American Philosophical Society** – Lewis and Clark Fund for Exploration and Research (PI): *Studying the effects of environmental change on the prehistoric settlement of Southwest Madagascar* \$5,000 2020-2021
- **NASA Pennsylvania Space Grant Consortium** – Graduate Research Fellowship (PI) [Grant No. NNX15AK06H] \$5,000 2019-2020

Internal Research Funding

- **Columbia University Provost's Teaching and Learning Grants:** Innovative \$10,950 2024-2025

Course Design (**co-PI**, PI Z. Crossland), *Collaborative learning through map-making*

- **Lamont-Doherty Earth Observatory Climate Center, CU** – Seed Grant (**PI**), \$13,230 2023-2024
Examining the role of subsistence economies on long-term socio-ecological dynamics on Madagascar
- **Department of Anthropology, PSU** – Hill Fellowship Post-Comprehensive Award (**PI**) \$5,000 2021
- **Liberal Arts Research and Graduate Studies Office, PSU** – Dissertation Support Grant (**PI**) \$1,994 2020-2021
- **Africana Research Center, PSU** – Research Grant (**PI**): \$1,000 2020-2021
An investigation of environmental change and its effects on settlement and mobility in Southwest Madagascar since the Late Holocene
- **African Studies Program, PSU** – Dickerson Family Fund Award (**PI**) \$900 2020
Evaluating Settlement Strategies in Southwest Madagascar Using Remote Sensing and Spatial Analysis
- **Department of Anthropology, PSU** – Hill Fellowship Pre-Comprehensive Award (**PI**) \$1,220 2020
- **Energy and Environmental Sustainability Laboratories/Laboratory for Isotopes and Metals in the Environment, PSU** – Graduate Student Seed Grant (**PI**): *Improved Carbon-14 Dating of Ratite Eggshell fragments from East Africa and Madagascar* \$1,735 2020
- **Institute for Cyber-Science, PSU** – Seed Grant (**Co-PI**, PI: Kristina Douglass): \$25,000 2019-2020
Machine Learning and the Preservation of Cultural Heritage on Madagascar
(I conceived of and wrote this grant largely independently, but graduate students could not serve as the PI)
- **Department of Anthropology, PSU** – Hill Fellowship Graduate Training Award (**PI**) \$1,500 2019
- **Energy and Environmental Sustainability Laboratories, PSU** – Graduate Student Seed Grant, PSU (**PI**): *Carbon-14 Dating of New Archaeological Sites in Southwest Madagascar: New Insights into the Early Settlement History of the Indian Ocean* \$2,430 2019

Academic Awards

- William T. Sanders Graduate Award in Anthropology \$729 2021
- 3rd Place Award in Social & Behavioral Sciences, 36th Annual Graduate Exhibition (PSU) \$100 2021
- Liberal Arts Teaching and Research Scholarship (PSU) \$1,000 2020
- Graduate Scholar Award (PSU) \$2,000 2020
- Graduate Scholar Award (PSU) \$2,000 2019
- Richard Antoun Faculty Award for Excellence in Anthropology \$250 2017

Travel Funding

- Conference Travel Award, Department of Anthropology (PSU) \$250 2022
- RGSO Conference Travel Award (PSU) \$400 2021
- RGSO Conference Travel Award (PSU) \$600 2019
- AGSA UPAC Travel Fund (PSU) \$604 2019

– RGSO Conference Travel Award (PSU)	\$400	2019
– Graduate Student Conference Travel Reimbursement Funding (BU)	\$400	2018
– Travel Grant, Department of Anthropology (BU)	\$500	2018
– Travel Grant, Department of Anthropology (BU)	\$500	2017
– Undergraduate Student Conference Travel Fund (BU)	\$300	2017
– Harpur Edge Student Support Fund (BU)	\$500	2017

NON-MONETARY AWARDS & RECOGNITION

– Top Cited Article in <i>Archaeological Prospection</i> in 2024	2026
– Top Cited Article in <i>Archaeological Prospection</i> between 2021-2022	2023
– Top Cited Article in <i>Archaeological Prospection</i> between 2019-2020	2021
– Top Downloaded Paper in <i>Archaeological Prospection</i> between 2018-2019	2020
– National Science Foundation Graduate Research Fellowship: Honorable Mention	2017

TEACHING INTERESTS

– Remote sensing, GIS, and spatial analysis	– Environmental archaeology	– Island & coastal studies
– Climate and Society	– Complex Adaptive Systems	– Environmental Modeling
	– Human behavioral ecology	– Archaeological and anthropological theory

TEACHING EXPERIENCE

Guest Lectures

2024	SDEV U6240 (Environmental Science for Sustainable Development), Graduate Seminar. Topic: Environmental Geoarchaeology: Using the Past to Inform the Present
2024	CLMT 6030 (Integrative and Justice-Centered Research in Environment and Climate), Graduate Seminar. Topic: Indigenous critiques of the “Anthropocene”
2022	ANTH209C (World Archaeology: Global Perspectives on the Past): <i>Satellites, machine learning, and community collaboration: A 21st century approach to environmental archaeology.</i> Washington University in St. Louis. 14 September.
2022	ANTH572 Graduate Seminar (Advances in Anthropological Methods): <i>Settlement Distribution and Land Use Patterns in Southwest Madagascar Since the Late Holocene.</i> Penn State. 23 March.

Courses Taught

Columbia University

2025	Anthropology 4066, Hydropolitics: GIS approaches to urbanism and the politics of water in Madagascar, Course Development, Instructor (Spring Term), 7 students.
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Penn State University

2022	Introduction to Geospatial Methods in Anthropological Research (Penn State), Course Development, Instructor (Summer term), 15 students
2022	ANTH 11: North American Archaeology, Instructor , Spring term [online format], 160 students

- 2021 ANTH 1: Understanding Humans (Penn State), **Instructor**, Summer and Fall term [online format], 100 students per class
- 2020-21 ANTH 140: Anthropology of Alcohol (Penn State), **Teaching Assistant** (FA-2020, SP-2021) [online format], 1,000 students (largest course offered by Penn State). [Duties included training undergraduate TAs, logistical planning of course materials, and holding office hours]
- 2020 ANTH 1: Understanding Humans (Penn State), **Teaching Assistant** (Spring Term), 100 students [Duties included grading assignments, holding office hours, and developing and presenting occasional lectures]
- 2019 ANTH 426W: Archaeological Laboratory Analysis (Penn State), **Teaching Assistant** (Spring term), 10 students. [Duties included teaching writing development skills, grading written assignments, holding office hours, and assisting students with lab-work]
- 2018-19 ANTH 9: Rise of Civilization in the Old World (Penn State), **Teaching Assistant** (Fall & Spring term), 130 students [Duties included presenting occasional lectures, leading review sessions, holding office hours, and grading assignments]

Advisory/Mentorship Activities
(PSU = Penn State, CU = Columbia University)

- Thesis/Dissertation Committees
 - Fetra Randriatsara (University of Cape Town, **External Examiner**)
- Graduate Student Mentorship

○ Clare Randolph (CU, Earth & Environmental Studies)	2022 – Present
○ Chiamaka Magnut (PSU, Anthropology; CU, Earth & Environmental Studies)	2021 – Present
○ Abiola Ibirogbra (PSU, Anthropology; CU, Earth & Environmental Studies)	2021 – Present
- Undergraduate Honors Thesis Advising

○ Claire Byrnes (PSU, Geography & Anthropology)	2021 – 2022
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- Undergraduate Research Assistants*

○ Sydney Ernst (PSU)	Spring 2021
○ Claire Byrnes (PSU)	2020 – 2021

* I trained these students in geospatial technologies and their applications to archaeology and cultural heritage research. This included satellite remote sensing and GIS analysis, basic coding in Javascript and R, predictive modeling, statistical machine learning, as well as data collection and storage practices.

INVITED TALKS

- 2025 *How can societies cope with rapid change? Seeking lessons at the Intersection of Geospatial and Archaeological Data Science.* Lecture presented to the Department of Data Science, University of Oregon, Eugene, OR, USA.

- 2024 *Exploring Histories of Socioecological Systems at the Intersection of Geospatial and Archaeological Science*. Lecture presented at the Department of Earth and Environmental Science Seminar Series, University of Illinois Chicago, USA.
- 2024 *Unraveling the Present and the Past: Remote Sensing at the Intersection of Environmental and Archaeological Science*. Lecture presented to the Department of Earth and Environmental Systems, Indiana State University, Terre Haute, IN, USA.
- 2024 *Community Engagement and Geospatial Archaeology: Lessons from Madagascar*. Lecture presented at the Geospatial Systems Centre for Doctoral Training Seminar Series, Newcastle University, U.K.
- 2024 *Humanizing the gaze of the machine: AI, anthropology, and the study of complex socioecological systems*. Lecture presented to the Department of Anthropology, Purdue University, West Lafayette, IN, USA.
- 2023 *Human-environment interactions in deep time: How humanity's past holds lessons for Earth's future*. Lecture presented at the Madagascar Biodiversity Center Seminar Series, Antananarivo, Madagascar.
- 2023 *Climate & Society: what archaeology can teach us about human responses to climate change*. Lecture presented at Columbia University Red Cross Event on Climate Change, Columbia University.
- 2023 *Decisions and Consequences: understanding the long-term role of human livelihoods on ecological systems*. Lecture presented at the Biology & Paleoenvironment Division Seminar Series, Lamont-Doherty Earth Observatory.
- 2023 *Legacies of Human Land Use and their Lessons for Resilience and Sustainability*. Lecture presented at the Complexity Science, Modeling and Sustainability Seminar Series, Columbia University.
- 2023 *Archaeology of sustainability: lessons about coastal land-use strategies from SW Madagascar*. Lecture presented at the Past Global Changes (PAGES) Q-MARE Seminar Series, Smithsonian Tropical Research Institute.
- 2022 *The impacts of pastoralism on landscape ecologies in Southwest Madagascar*. Lecture presented at the Archaeological Sciences in Eastern Africa: The Great Conversation Q&A forum and lecture series, Department of Archaeology, Max Plank Institute for the Science of Human History. Jena, Germany.
- 2022 *Power-Dynamics and pathways towards equity in Landscape Research: a case study from archaeology*. Guest Lecture presented at the Decolonizing Water Scholarship Reading Group. The Pennsylvania State University, USA.
- 2021 *Settlement and Land Use in Late Holocene Southwest Madagascar*. Lecture presented at the Archaeological Science Seminar Series. Newcastle University, UK.

CONFERENCE PRESENTATIONS

(*Invited; 1. Undergraduate Student; 2. Graduate Student)

- 2027 Horsely, T., **Davis D.**, Barrier, C., Robin, B., Henry, E. "What is a house anyway? Using text-based AI models to identify cultural features in geophysical datasets." Paper to be presented at the 92nd Annual Meeting of the Society for American Archaeology, Indianapolis, IN.
- 2027 **Davis D.**, DiNapoli, R., Lipo, C. "Toward Label-Free Archaeological Prospection: Zero-Shot Detection and AI Interpretation of Earthworks from LiDAR in the United States." Paper to be presented at the 92nd Annual Meeting of the Society for American Archaeology, Indianapolis, IN.
- 2026 Alders, W., Dumitru, I., Mjema, E., Lim, J., **Davis, D.** "Inundation innovations: comparing datasets for modeling monsoonal floods, natural catastrophes, and long-term settlement reorganization on the Swahili Coast". Computer Applications in Archaeology Conference, Vienna, Austria. In "Computational Archaeology at Scale: Large Spatial Datasets for Unveiling Cultural Landscapes.
- 2026 Alders, W., Lim, J., Brunner, L., **Davis, D.** "Tree of the African Anthropocene: Remote sensing and landscape archaeology approaches to human-baobab entanglements". 9th International Landscape Archaeology Conference, Bamberg, Germany. In "The Future of (Landscape) Archaeology in the Anthropocene".
- 2025 Crossland, Z., **D.S. Davis**, E. Glass, S. Duncan², W. Falcon², F. Germer¹, S. Hou¹, G. Luo¹, R. Mouaouia¹, J. Switzer², J.S. Lim, C. Radimilahy, B. Rasoarifetra, N. A. Nasandrata², F. O. Rajaornarison², N. A. Rakotondrazafy², M. Rakotoniaina², F. E. Ravokatra², M. F. Razafiarison². "Mapping threats to heritage and archaeology in highland Madagascar". Poster to be presented at the 27th Biennial Annual Meeting of the Society of Africanist Archaeologists, Algarve, Portugal
- 2025 Mangut, C.² and **D.S. Davis** "High-Resolution Satellite Imaging for Archaeological Reconnaissance and Environmental Reconstruction: A Case Study of Dutsen Kura Hill, Jos Plateau, Nigeria". Paper to be presented at the 27th Biennial Annual Meeting of the Society of Africanist Archaeologists, Algarve, Portugal
- 2025* **Davis, D.S.** and K. Douglass "Surviving hypervariable socioecological conditions: A case from southwest Madagascar". Paper to be presented at the 27th Biennial Annual Meeting of the Society of Africanist Archaeologists, Algarve, Portugal
- 2025 Wolverton, S., C. Otaola, R. Figueroa, & **D. S. Davis**. "A Theoretical Framework for Action Ethnobiology from Historical Ecology and Environmental Justice Studies in the Context of Andean Puestero Pastoralism.". Paper presented at the 46th Annual Conference of the Society of Ethnobiology at Lake Tahoe, Nevada
- 2025* **Davis, D. S.**, W. J. D'Andrea, & K. Douglass "A Geoarchaeological Approach to Coupled Human-Natural Systems on SW Madagascar". Paper presented at the 89th Annual Meeting of the Society for American Archaeology, Denver, CO
- 2024* **Davis, D. S.** "Using Remote Sensing Archaeology to Assess Land Use Histories and Resilience". Paper presented at NASA and Archaeology from Space: A Symposium in Honor of Dr. Thomas L. Sever. 18-19 September, Huntsville, AL.

- 2024 Pham, K. V.¹, M. C., Andriambelomanana, M. C., E. Chevallier, F. Fenomanana, M. Ramananjanahary, R. Razafimagnefa, G. Manahira, **D. S. Davis**, K. Douglass, and S. J. Ivory. "Influences on past survival of the *Adansonia grandidieri* baobab in Velondriake, Southwest Madagascar." Poster presented at the Association for Tropical Biology and Conservation Meeting, Kigali, Rwanda
- 2024 **Davis, D. S.** and K. Douglass. "Traditional subsistence economies on Southwest Madagascar have long-term impacts on ecological productivity." Paper presented at the 89th Annual Meeting of the Society for American Archaeology, New Orleans, LA.
- 2023 **Davis, D. S.** "Legacies of human land use and their lessons for resilience and sustainability." Presented at the Columbia University Postdoctoral Research Symposium, 27 September.
- 2023 Pham, K. V.¹, M. C., Andriambelomanana, M. C., E. Chevallier, F. Fenomanana, M. Ramananjanahary, R. Razafimagnefa, G. Manahira, **D. S. Davis**, K. Douglass, and S. J. Ivory. "Ecological interactions between humans and baobabs in Velondriake, Southwest Madagascar." Paper presented at the Geological Society of America Connects 2023, 15-18 October, Pittsburgh, PA. DOI: 10.1130/abs/2023AM-393248
- 2023* **Davis, D. S.** and K. Douglass. "Behavioral Ecology and Evolutionary Approaches to Human-Environment Dynamics on SW Madagascar." Paper presented at the 88th Annual Meeting of the Society for American Archaeology, Portland, OR.
- 2023 Cook Hale, J. W., **D. S. Davis**, M. C. Sanger, and N. L. Hale. "Evaluating the use of bathymetric LiDAR for submerged archaeological site prospection." Poster presented at the 88th Annual Meeting of the Society for American Archaeology, Portland, OR.
- 2022* **Davis, D. S.** "'Big Data' and Archaeological Studies of Human-environmental dynamics." Invited Lightning Presentation and Roundtable Discussion at the Archaeological Institute of America, 2022
- 2022 **Davis, D. S.**, D. C. Buffa, T. Rasolondrainy, E. Creswell, C. Anyanwu, A. Ibiroga, C. Randolph, A. Ouarghidi, L. N. Phelps, F. Lahiniriko, Z. M. Chrisostome, G. Manahira and K. Douglass. "Bird's Eye Ethics: The panopticon and archaeological remote sensing." Poster presented at Frontiers in Archaeological Sciences 3: Rethinking the Paradigm, 7-9 October, Cornell University, Ithaca, NY.
- 2022 Cook Hale, J. W., **D. S. Davis**, M. Sanger, K. Woo, N. L. Hale, and C. A. Hemmings. "Remote sensing, machine learning, and the search for shallowly submerged formerly coastal sites: a case study from Apalachee Bay, Gulf of Mexico, Florida, U.S.A." Paper presented at the 2022 AIMA Conference: Decolonising Maritime Archaeology, Heritage and Museums in Australasia, 21-23 September, Adelaide, South Australia.
- 2022 **Davis, D. S.**, T. Rasolondrainy, G. Manahira, V. Andriankaja, L. Hubertine, R. Justome, F. Lahiniriko, H. Léonce, R. Roi, F. Victorian, M. B. J. Clovis, V. Voahirana, T. L. Carina, A. J. Yves, Z. M. Chrisostome, B. Manjakahery, and K. Douglass. "Social networks as risk-mitigation strategies on Southwest Madagascar: A multiproxy analysis." Paper presented at the 16th Congress of the Pan African Archaeological Association for Prehistory and the Related Studies, 2022, Zanzibar, Tanzania.

- 2022* **Davis, D. S.**, C. Howe, H. Theiss, L. N. Phelps, C. Anyanwu, R. Opitz, C. Klehm, and K. Douglass. "Elucidating the pastoralist niche on SW Madagascar using automated remote sensing." Paper presented at the 87th Annual Meeting of the Society for American Archaeology, 2022, Chicago.
- 2022 Randolph, C.¹, **D.S. Davis**, D. Buffa, and K. Douglass. "Collaborative Archaeology in SW Madagascar: Implications for Conservation using Direct Stakeholder Involvement." Paper presented at Synergy Between Palaeo-scientists and Stakeholders for Biodiversity Conservation in Madagascar and its Surrounding Islands Workshop, 23-24 March. Virtual.
- 2021* **Davis, D. S.**, G. Caspari, C. P. Lipo, and M. C. Sanger. "Using Deep Learning to Detect Rare Archaeological Features: A Case from Coastal South Carolina, USA." Paper presented at the International Meeting for Applied Geoscience & Energy Meeting. Session Title: Applications for Archaeology, Void, and Target Detection. Denver, CO, 26 September – 1 October.
- 2021* **Davis, D. S.** "Documenting the Ecological Legacy of Land-Use by Small-Scale Communities in Southwest Madagascar." Paper presented at the UK Earth Observation Week - Supporting climate and environmental sustainability through Earth Observation, 2021, Virtual.
- 2021 **Davis, D. S.**, V. Andrianakaja, T. L. Carina, Z. M. Chrisostome, C. Colombe, F. Fenomanana, L. Hubertine, R. Justome, F. Lahiniriko, H. Léonce, G. Manahira, B. V. Pierre, R. Roi, P. Soafiavy, F. Victorian, V. Voahirana, B. Manjakahery, and K. Douglass. "Remote Sensing and Spatial Analysis Reveal Coastal Settlement Strategies in Southwest Madagascar." Paper presented at the 25th Society of Africanist Archaeologists, Oxford, UK, 14-15 August.
- 2021* **Davis, D. S.**, L. N. Phelps, C. Anyanwu, and K. Douglass. "Grasslands and Pastoralists: Palaeoecological, Archaeological, and Geophysical Evidence. Lightning Talk presented at the Malagasy Grassy Biomes Workshop, Virtual, 27-30 July.
- 2021* **Davis, D. S.** "Using machine intelligence to locate ephemeral archaeological landscape modifications: A case from Madagascar." Paper presented at CAA 2021: Digital Crossroads. Session title: From Artificial Intelligence to Stratigraphic Reality. Dynamics of an Inverse Process for AI Applications in Archaeology. 14-18 June.
- 2021 **Davis, D. S.** "Archaeology Highlights Strategies for Coping with Climate Change." Oral presentation at the 36th Annual Graduate Research Exhibition, 24-26 March, Pennsylvania State University.
- 2020 **Davis, D. S.** "Quantifying Ancient Landscape Modifications using Machine Learning and Evolutionary Theory: A Case Study from Madagascar." Poster presented at the Center for Computational Mathematics and Applications Workshop on Mathematical Machine Learning and Application. University Park, PA. 14-16 December.

- 2020 **Davis, D. S.**, and K. Douglass. "Archaeological Settlement on Madagascar Revealed Using Semi-Automated Satellite Remote Sensing. Poster presented at the Institute for Computational Data Sciences (ICDS) Symposium, Pennsylvania State University. 21-22 October.
- 2020 Douglass, K., and **D. Davis**. "Archaeology of Adaptation and Mobility." Lightning Talk presented at Microsoft's AI for Earth Digital Summit. 25-26 June.
- 2019 **Davis, D. S.** "Developing a Predictive Model for Locating Cultural Deposits on Southwest Madagascar." Paper presented at the 14th Madagascar Workshop, University of Toronto, 25-26 October.
- 2019 **Davis, D. S.**, M. C. Sanger, and C. P. Lipo. "Shell Rings and Settlement Organization in the Coastal American Southeast: New insights from remotely sensed data." Poster presented at the 84th Annual Meeting of the Society for American Archaeology, Albuquerque, NM, 10-14 April.
- 2018 **Davis, D. S.** "Unveiling the Past: Using LiDAR to Discover Lost Archaeological Features in the American Southeast." Poster presented at the 3rd Annual GIS Day, Binghamton University, 16 February. <https://orb.binghamton.edu/gisday/1>.
- 2018 **Davis, D. S.** "Filling in the Map: Object-Based Image Analysis and the Identification of Shell Rings on Hilton Head Island, SC." Poster presented at the 83rd Annual Meeting of the Society for American Archaeology, Washington D. C., 11-15 April.
- 2017 **Davis, D.** and C. Lipo. "The Benefits of Short-Wave Infrared Imagery for Archaeological Landscape Analysis: A Case Study from Easter Island, Chile." Poster presented at the 82nd Annual Meeting of the Society for American Archaeology. Vancouver, BC, 29 March – 2 April.
- 2017 **Davis, D.** and C. Lipo. "Resource Scarcity and Monumental Architecture: Cost Signaling on Rapa Nui (Easter Island), Chile." Poster presented at Binghamton Research Days. Binghamton University, 21 April.

PUBLISHED DATASETS

- 2024 Agiadi, K., M. Bas Lopez, J. D. Cybulski, **D. S. Davis**, D. Alunni, K. K. T. Aye, G. Boretto, G. Carenti, D. Chattopadhyay, M. Coll, A. Colonese, A. Dahakey, E. Dillon, K. Douglass, N. Garcia Chapori, P. N. Lampri, C. Laprida, D. Leal, C.-H. Lin, ... F. Ten Brink. "Q-MARE database on pre-industrial climate and human impacts on marine ecosystems". Zenodo. <https://doi.org/10.5281/zenodo.12752329>
- 2023 **Davis, D. S.**, R. J. DiNapoli, G. Pakarati, T. L. Hunt, and C. P. Lipo. "Data and Code for 'High Resolution short-wave infrared imagery improves estimates of ancient rock gardening distribution on Rapa Nui (Easter Island)'." Github. https://github.com/d-davis/rapanui_mulching.
- 2022 Quintus, S., **D. S. Davis**, and E. E. Cochrane. "Supporting Data – Evaluating Mask R-CNN Models to Extract Terracing across Oceanic High Islands: an example from Sāmoa." Zenodo. <https://doi.org/10.5281/zenodo.7312103>.

- 2022 **Davis, D. S.**, T. Rasolondrainy, G. Manahira, S. W. Hixon, V. Andriankaja, L. Hubertine, R. Justome, F. Lahiniriko, H. Léonce, R. Roi, F. Victorian, M. B. J. Clovis, V. Voahirana, T. L. Carina, A. J. Yves, Z. M. Chrisostome, B. Manjakahery, and K. Douglass. "Evidence for Extensive Social Networks as Risk-Mitigation Strategies on Southwest Madagascar." Github. https://github.com/d-davis/mada_ceramic.
- 2021 **Davis, D. S.**, and J. Lundin. "Locating Charcoal Production Sites in Sweden Using Hydrological Algorithms and Deep Learning." Penn State ScholarSphere. <https://doi.org/10.26207/zfc5-rd31>.
- 2021 **Davis, D. S.**, and K. Douglass. "Remote Sensing Reveals Lasting Legacies of Land-Use by Small-Scale Communities in the southwestern Indian Ocean. Penn State ScholarSphere. <https://doi.org/10.26207/zmsr-tc92>.
- 2021 **Davis, D. S.**, G. Caspari, C. P. Lipo, and M. C. Sanger. "Deep Learning Reveals Extent of Archaic Native American Shell Ring Building Practices." Penn State ScholarSphere. <https://doi.org/10.26207/ry6k-q463>.
- 2019 **Davis, D. S.**, V. Andriankaja, T. L. Carina, Z. M. Chrisostome, C. Colombe, F. Fenomanana, L. Hubertine, R. Justome, F. Lahiniriko, H. Léonce, G. Manahira, B. V. Pierre, R. Roi, P. Soafiavy, F. Victorian, V. Voahirana, B. Manjakahery, and K. Douglass. "Satellite-based remote sensing rapidly reveals extensive record of Holocene coastal settlement on Madagascar. Penn State ScholarSphere. <https://doi.org/10.26207/1a47-pw11>.
- 2019 **Davis, D. S.**, R. J. DiNapoli, M. C. Sanger, and C. P. Lipo. "The integration of lidar and legacy datasets provides improved explanations for the spatial patterning of shell rings in the American Southeast." Penn State ScholarSphere. <https://doi.org/10.26207/7erq-3662>.
- 2018 **Davis, D. S.**, C. P. Lipo, and M. C. Sanger. "A comparison of automated object extraction methods for mound and shell-ring identification in coastal South Carolina." Anthropology Datasets 4. https://orb.binghamton.edu/anthro_data/4/.
- 2018 Lipo, C. P., M. Sanger, and **D. Davis**. "Automated mound detection using LiDAR and Object-Based Image Analysis in Beaufort County, SC." Anthropology Datasets 3. https://orb.binghamton.edu/anthro_data/3.

CARTOGRAPHIC WORK

- 2020 Figure 2: "Teotihuacan and other Terminal Formative and Classic period sites mentioned in Central Mexico" in K. G. Hirth, D. Carballo, and B. Arroyo (eds.) *Teotihuacan: The World Beyond the City*. Dumbarton Oaks, Washington D.C.
- 2020 Figure 9-1: "Five different periodic rendezvous or trade fairs held in the western United States before European contact." In K.G. Hirth *The Organization of Ancient Economies: A Global Perspective*.

2020 Figure 9-7: “Merchant circuits between large and small markets across three different environmental zones.” In K.G. Hirth *The Organization of Ancient Economies: A Global Perspective*.

FIELDWORK

- July – August 2023 – Coral Coring Surveys and Paleoclimate Reconstruction, SW Madagascar: **Co-PI** (with K. Douglass, J. Zinke, and T. Rasolondrainy)
- April 2021 – July 2021 – Archaeological Excavations and Paleoecological Survey at Velondriake, Madagascar: **Principal Investigator** [planned and organized field logistics, I was not in the field due to the COVID-19 Pandemic]
- May 2020 – November 2020 – Archaeological Survey of the Velondriake Region, Morombe, Madagascar: **Principal Investigator** [planned and organized field logistics, I was not in the field due to the COVID-19 Pandemic]
- July 2019– August 2019 – Terrestrial and Marine Archaeological Survey of the Velondriake Region, Morombe, Madagascar: **Principal Investigator**
- July 2018 – Sea Pines Archaeological Research Project, Hilton Head Island, South Carolina, USA: **Field Archaeologist** (PI: Matthew Sanger)
- October 2017 – Archaeological Survey in Beaufort County, South Carolina, USA: **Principal Investigator**
- July 2017 – August 2017 – Sea Pines Archaeological Research Project, Hilton Head Island, South Carolina, USA: **Field Archaeologist** (PI: Matthew Sanger)
- June 2015 – August 2015: The Gabii Project, Rome, Italy: **Student** (PIs: Nicola Terrenato, Marcello Mogetta)

LABWORK

- August 2023 – Present: **Postdoctoral Researcher** in the Lipid Biomarker and Organic Geochemistry Lab, Columbia University
- July 2022 – Present: **Postdoctoral Researcher** in the Olo Be Taloha Lab, Columbia University
- June 2021 – August 2021: **Graduate Research Assistant** in the Olo Be Taloha African Environmental Archaeology Lab, Pennsylvania State University
- June 2020 – August 2020: **Graduate Research Assistant** in the Primate Functional Morphology Lab, Pennsylvania State University
- January 2019 – May 2022: **Doctoral Researcher** in the Olo Be Taloha African Environmental Archaeology Lab, Pennsylvania State University
- March 2019 – May 2022: **Doctoral Researcher** in the Isotope and Paleoclimatology Lab, Pennsylvania State University
- September 2018 – September 2020: **Doctoral Researcher** in the Human Environmental Dynamics Lab, Pennsylvania State University
- February 2017 – May 2018: **Lab Assistant** in the Archaeological Imaging, Mapping, and Morphology Lab, Binghamton University
- February 2017 – May 2018: **Graduate Researcher** in the FRI Remote Sensing Lab, Binghamton University

TECHNICAL TRAINING

Methods	Software/Technology
GIS, spatial analysis, remote sensing, image analysis, predictive modeling, 3D-mesh creation, object-based image analysis, deep learning, photogrammetry, geophysical survey, archaeological survey, archaeological excavation, data processing and analysis, ceramic analysis, acid hydrolysis for AMS dating preparation, Bayesian Statistics	ArcGIS, QGIS, SAGA, Whitebox GAT, ENVI, eCognition, R, Surfer, GPR, Magnetometer, Python, MeshLab LiDAR, multispectral imagery, thermal imagery, SAR

WORKSHOPS ATTENDED

- 2025: *Megafauna Depletion and Holocene Ecological Networks*. Working Group organized by K. Harper and C. Hofman. Santa Fe Institute, Santa Fe, NM.
- 2023: Winter Workshop on Complex Systems 2023, Institute for Advanced Studies, Amsterdam, NL.
- 2020: DataArc Workshop: Tool Evaluation. Center for Advanced Spatial Technologies, University of Arkansas, November 16.
- 2020: COMPASS Science Communication Training Workshop, The Pennsylvania State University
- 2017: Using R for Archaeological Data Analysis, Mapping, and Visualization Workshop. 82nd Annual Meeting of the Society for American Archaeology Meeting, Vancouver, BC.

ADDITIONAL TRAINING

- 2013 Advanced Open Water Diving Certification (SSI)
 – Advanced certifications: Wreck, Night/limited visibility, Deep, Drysuit, Nitrox (EAN 40%)
- 2012 Open Water Diving Certification (SSI)

PROFESSIONAL MEMBERSHIPS

- | | |
|---|----------------|
| – Society for Africanist Archaeologists | 2019 – Present |
| – Sigma Xi (Associate Member 2018-2024; Full Member since 2024) | Inducted 2018 |
| – Society for American Archaeology | 2016 – 2026 |
| – Phi Beta Kappa Honor Society | Inducted 2017 |
| – Gamma Theta Upsilon, International Geographical Honor Society | Inducted 2017 |

ACADEMIC SERVICE

Organized Workshops/Symposia

- 2022 Pan-African Archaeological Association Conference 2022 Session Title: *Approaches to studying African Landscapes* (**co-organizer**)
- 2021 Malagasy Grassy Biomes Working Group (**co-organizer**)

Editorial Positions

2025 – Present Co-Editor in Chief, ***Archaeological Prospection***
 2022 – 2025 Guest Editor, Special issue of ***Plants, People, Planet***
 2020 – 2025 Early Career Researcher Board, ***Archaeological Prospection***
 2020 – 2021 Guest Editor, Special Issue of ***Archaeological Prospection***

Review Panels:

2025: National Science Foundation (NSF) – Social, Behavioral, and Economic Sciences (SBE) Division.

External Reviewer:

Grant Agencies: NASA, Graduate Women in Science (GWIS) National Fellowship Program, National Geographic Society, The Leakey Foundation

Books: *University Press of Florida*

Journals:

ARCHAEOLOGY/ANTHROPOLOGY: *Journal of Archaeological Science, American Antiquity, Heritage Science, Heritage, Antiquity, Journal of Archaeological Research, African Archaeological Review, Journal of Archaeological Science: Reports, Journal of Field Archaeology, Advances in Archaeological Practice, Canadian Journal of Archaeology, Southeastern Archaeology, Geoarchaeology, Current Anthropology, Journal of Island and Coastal Archaeology, Journal on Computing and Cultural Heritage, Journal of Cultural Heritage, Journal of Archaeological Method and Theory, Journal of Maritime Archaeology*

GEOGRAPHY/GEOSCIENCES: *Archaeological Prospection, Remote Sensing of Environment, Remote Sensing, ACME: An International Journal for Critical Geographies, International Journal of Applied Earth Observation and Geoinformation, Geosciences, IEEE Sensors Journal, The Journal of Maps*

ENVIRONMENTAL STUDIES: *The Holocene, Quaternary International, Land, Frontiers in Environmental Archaeology, Earth Science Informatics*

PLANT SCIENCES: *Plants, People, Planet*

MULTIDISCIPLINARY: *Nature Communications, Scientific Data, PLOS ONE, The Innovation, Water, Sustainability, Royal Society Open Science, Information, Global Sustainability, Scientific Reports*

Book Chapters: *Analecta Praehistorica Leidensia (APL), Max Planck Institute*

Departmental Service

2024 Postdoctoral-Grad Student Mentoring Event Participant, Lamont-Doherty Earth Observatory, Columbia University

2021 – 2022 Anthropological Research in Science Education (ARISE) Committee (Summer program for historically underrepresented undergraduate students from Penn State Campuses)

2020 – 2022 Graduate-Undergraduate Mentorship Program

2020 – 2021	Anti-Racism Pedagogy Group Member (Part of an effort to design an Anthropology of Race and Racism course for undergraduate students, PSU)
2019 – 2021	Member of Colloquium Committee, Department of Anthropology (PSU)
2019 – 2022	Member of Outreach Committee, Department of Anthropology (PSU)
2019 – 2021	Member of international effort to increase impact of student scholarship (International Journal of Student Research in Archaeology)
2019 – 2022	Member of Awards Committee, Department of Anthropology (PSU)
2019 – 2021	Mentor for 3 first-year PhD students in Anthropology Department (PSU)
2014 – 2015	Editor for undergraduate academic journal at Binghamton University

MEDIA COVERAGE

(* indicates newsletter research highlight)

- Spring 2026, **American Archaeology** (Vol. 30 No. 1), “When Data Outruns the Dig”. pp. 33-40.
- September 11, 2024, **AFP News (Paris)**, “‘Ecocide’ On Easter Island Never Took Place, Studies Suggest”
- July 31, 2024, **EcoAméricas**, “New research, and a new narrative, for Easter Island”
- June 27, 2024, **Bloomberg**, “Easter Island Collapse Gets the Fresh Look It Deserves”
- June 26, 2024, **Advanced Science News**, “Study challenges theory of population collapse on Easter Island”
- June 21, 2024, **The Guardian**, “Easter Island study casts doubt on theory of ‘ecocide’ by early population”
- June 21, 2024, **The Independent**, “Popular theory claiming Easter Island’s population collapsed due to ‘ecocide’ is disproved”
- June 21, 2024, **Science News**, “A new study challenges the idea that Rapa Nui islanders caused an ‘ecocide’”
- June 21, 2024, **CNN**, “Satellite imagery may provide a missing puzzle piece in Easter Island saga”
- June 21, 2024, **Popular Archaeology**, “Rapa Nui community had fewer members than thought, making overpopulation and ecological collapse unlikely” [COVER STORY]
- June 21, 2024, **L’ADN**, “Non, les Rapa Nui n'ont pas provoqué de désastre écologique sur l'île de Pâques”
- June 21, 2024, **Live Science**, “Easter Island's population never collapsed because it never got that big, researchers suggest”
- June 21, 2024, **Superinteressante**, “Estudo sobre Ilha de Páscoa põe em xeque tese de que povo rapanui fez ecocídio”
- November 27, 2023, **Science News**, “Reindeer herders and scientists collaborate to understand Arctic warming”
- May 18, 2022, **Phys.org**, “Tree loss on Madagascar not caused by small-scale fires used for land clearing”
- May 18, 2022, **EurekAlert**, “High rates of landscape degradation not product of landscape fire”
- November 24, 2021, **Geospatial Solutions**, “GIS helps archaeologists trace a mother’s journey”

- October 24, 2021, **Science Alert**, [“Satellite Data Shows Even Small Ancient Gatherer Communities Shaped The Landscape”](#)
- October 19, 2021, **EurekAlert**, **Phys.org**, **ScienceDaily**, **Mirage News**, **Eurasia Review**, **Bioengineer.org**, **Fuentitech**, **Florida News Times**, **TechCodex**, **NewsBinding**, “Small-scale foragers left more than footprints on the landscape”
- October 7, 2021, **ZME Science**, [“Machine learning reveals archaeology from up to 5,000 years ago”](#)
- August 30, 2021, **Archaeology Magazine**, [“Archaic Shell Rings Spotted in Southeastern United States”](#)
- August 20, 2021, **AZO Robotics**, [“Researchers Use AI and Machine Learning to Reveal Undiscovered Shell Rings”](#)
- August 19, 2021, **EurekAlert**, **Phys.org**, **Mirage News**, **Penn State News**, Remote sensing and machine learning reveal Archaic shell rings
- July 21, 2021, **Penn State News**, **Phys.org**, **Archaeology News Network**, [“The ethics of remote sensing in archaeology”](#)
- January 22, 2021, **Medium**, [“Archaeologists Use AI to ‘Dig Deeper’”](#)
- May 14, 2020, **Unite.AI**, [“AI Is Dramatically Changing Archaeology, Discovering New Sites And Artifacts”](#)
- May 7, 2020, **Singularity Hub**, **The Mandarin**, [“The New Indiana Jones? AI. Here’s How It’s Overhauling Archaeology.”](#)
- February 25, 2020, **Futurity**, [Ancient Madagascar Shows Humans Make New Places Suit Them](#)
- February 18, 2020, **Penn State News**, [Computers scour satellite imagery to unveil Madagascar’s mysteries](#)
- *Davis, Dylan. 2019. Locating and Protecting Cultural Heritage using Remote Sensing Survey in Madagascar. *The Current: Newsletter of the Island & Coastal Archaeology Interest Group*. 7(1):13.
- July 27, 2018, **Azo Sensors**, [New Image Analysis Method Used by Archaeologists to Identify Old North American Mounds](#)
- July 23, 2018, **EurekAlert**, **Science Daily**, **Phys.org**, **Heritage Daily**, **ScienMag** “Archaeologists identify ancient North American mounds using new image analysis technique”
- February 19, 2018, **BU Pipe Dream** [Geography department holds third annual GIS Day](#)
- January 22, 2018, **BU Pipe Dream** [BU research challenges Easter Island’s historical narrative](#)
- July 14, 2017, **The Island Packet**, [What’s buried in Sea Pines and more than 4,000 years old? Find out who’s digging for answers](#)

PUBLIC OUTREACH ACTIVITIES

- 2024 **Volunteer** at Lamont-Doherty Earth Observatory’s Open House Event, Palisades, NY
- 2024 **Interview** for *Weekend*, BBC World Service Radio (June 29, 2024)
- 2024 **Interview** for the Saturday Morning Podcast, Radio New Zealand (June 29, 2024), [“Easter Island 'ecocide' theory challenged”](#).
- 2021 **Invited Guest Speaker** on ArchaeoTech Podcast, Ethics and Remote Sensing Archaeology – Episode 161 (<https://www.archaeologypodcastnetwork.com/archaeotech/161>)

- 2020 Graduate School Application Preparation Workshop, Penn State, **Co-Organizer**
- 2020, Talked to elementary schoolers about African Archaeology, Germantown Academy, PA,
2021 **Invited Speaker**
- 2020 ENVISION Workshop, Eberly College of Science, Penn State University, **Volunteer**
- 2019 – Community Education Extended Learning (CEEL) Program, State College (Pennsylvania) School
2020 District, **Volunteer**
- 2019 Haunted-U (Eberly College of Science, Penn State University), **Volunteer**
- 2018 **Invited Guest Speaker** on ArchaeoTech Podcast, Using Math and Maps to find Mounds in the Southeast US – Episode 86 (<https://www.archaeologypodcastnetwork.com/archaeotech/86>).